

HUNDRED DAYS OF CODE

DAY 11

QUESTION: Write a menu-driven C program using switch-case to input a positive integer and find the sum of squares of its digits using a while loop. **E.g.** Number: 123, Sum=14.

STEP 1: Required input, and output

INPUT: A positive integer n.

OUTPUT: Sum of squares of all the digits of the given number.

STEP 2: ALGORITHM

1. Start
2. Declare variables n, digit, and sum.
3. Input a positive integer n.
4. Initialize sum = 0.
5. Repeat the following steps while $n > 0$:
 - Find the last digit using $\text{digit} = n \% 10$.
 - Calculate the square of the digit.
 - Add the square to sum.
 - Remove the last digit using $n = n / 10$.
6. Display sum.
7. Stop.

STEP 3: PSEUDOCODE

START

INPUT n

sum $\leftarrow$ 0

WHILE $n > 0$ DO

 digit $\leftarrow$ n MOD 10

 sum $\leftarrow$ sum + (digit $\times$ digit)

 n $\leftarrow$ n DIV 10

END WHILE

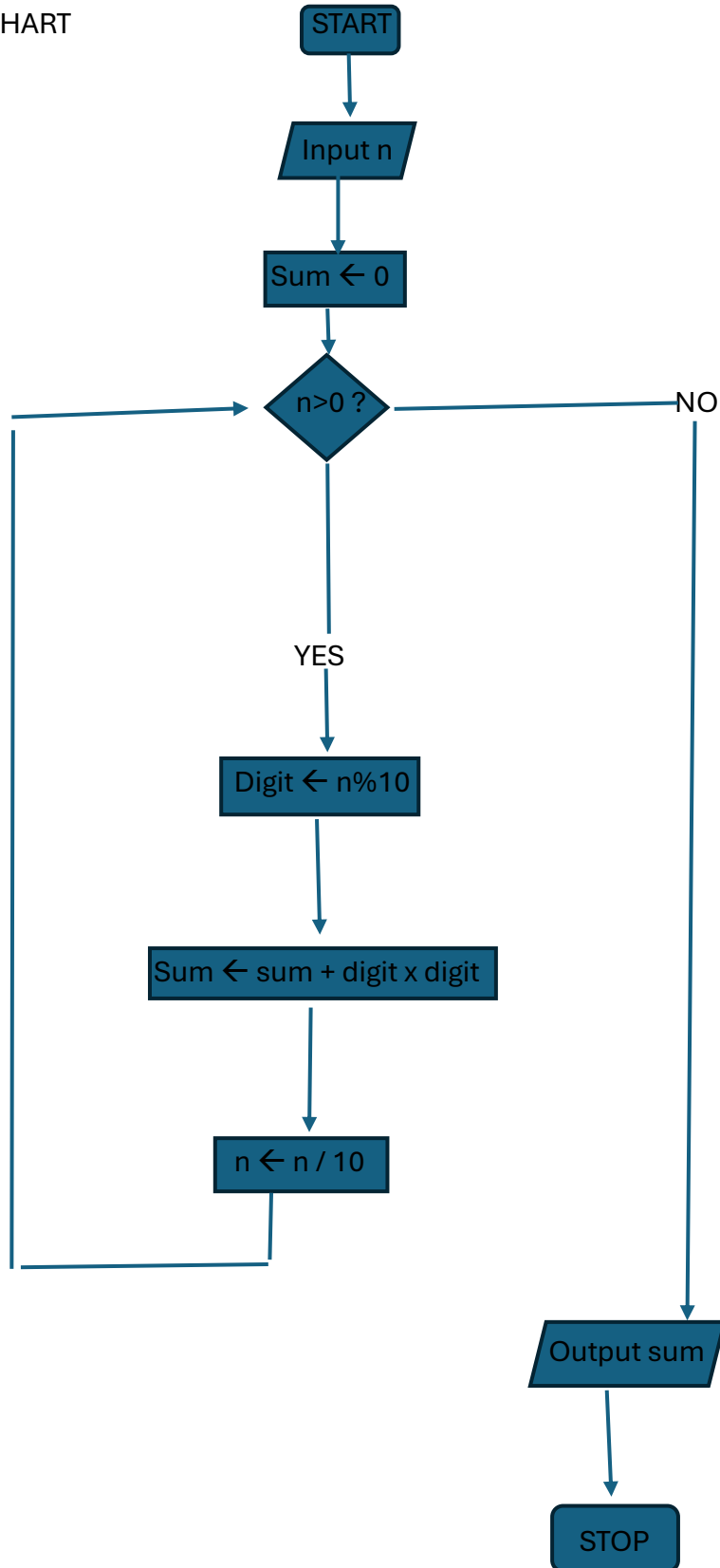
OUTPUT sum

STOP

HUNDRED DAYS OF CODE

DAY 11

STEP 4: FLOWCHART



HUNDRED DAYS OF CODE

DAY 11

STEP 5: REQUIRED C PROGRAM:

```
#include <stdio.h>

int main()
{
    int n, digit, sum = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while (n > 0)
    {
        digit = n % 10;
        sum = sum + digit * digit;
        n = n / 10;
    }

    printf("Sum of squares of digits = %d\n", sum);

    return 0;
}
```

STEP 6: TEST CASE TABLE

TEST CASE	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
1	123	14	14	PASS
2	100	1	1	PASS
3	12	5	5	PASS

HUNDRED DAYS OF CODE

DAY 11

STEP 6: COMPILATION AND EXECUTION

```
(kali㉿kali)-[~]  
$ vi sum.c  
  
(kali㉿kali)-[~]  
$ cc sum.c -o sum  
  
(kali㉿kali)-[~]  
$ ./sum  
Enter a positive integer: 123  
Sum of squares of digits = 14  
  
(kali㉿kali)-[~]  
$ ./sum  
Enter a positive integer: 100  
Sum of squares of digits = 1  
  
(kali㉿kali)-[~]  
$ ./sum  
Enter a positive integer: 12  
Sum of squares of digits = 5  
  
(kali㉿kali)-[~]  
$
```

.....